

FinMA-RAG: A Multi-Agent Retrieval and Reasoning for Financial Reports

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Abstract

Answering questions over full SEC filings is challenging because relevant evidence is dispersed across long, irregular, and table-heavy documents, conditions under which standard retrieval-augmented generation (RAG) pipelines routinely fail. We present FinMA-RAG, a domain-adaptive system that integrates structure-preserving ingestion, hybrid sparse-dense retrieval with LLM-based query expansion, and a multi-agent workflow that delegates all numerical operations to explicit tools. On the full FinanceBench benchmark, FinMA-RAG improves GPT-4o-judged accuracy from 19.3% to 44.0%. These results show that robust financial QA requires jointly addressing structural noise, terminology drift, and numerical grounding through an integrated retrieval-and-reasoning pipeline, rather than relying solely on stronger retrievers or larger language models.

1 Introduction

Financial question answering over SEC filings poses challenges far beyond those seen in typical long-document QA. Relevant evidence is dispersed across multi-hundred-page narratives, table-heavy financial statements, and cross-referenced footnotes. Correct answers often require resolving issuer-specific terminology, aligning fiscal periods, and performing precise numerical operations, capabilities that standard retrieval-augmented generation (RAG) pipelines do not reliably provide.

Dense retrievers frequently surface semantically related but period-mismatched disclosures, whereas sparse retrievers fail under lexical variation such as “revenue” versus “net sales.” Even when the relevant text is retrieved, large language models (LLMs) may hallucinate arithmetic, misread multi-year tables, or produce inconsistent numerical comparisons. Recent evaluations such as FinanceBench attribute these failures not to language modeling alone, but to the interaction of

three factors: structural fragmentation introduced during PDF ingestion, terminology drift during retrieval, and unreliable implicit numerical reasoning.

We introduce FinMA-RAG, a domain-adaptive framework designed to address these challenges in a unified manner. The system consists of: (i) structure-preserving ingestion that maintains table boundaries and document hierarchy; (ii) hybrid sparse-dense retrieval with LLM-driven query expansion and late-interaction reranking to improve both coverage and period specificity; and (iii) a multi-agent workflow where explicit tools perform all arithmetic to ensure verifiable numerical grounding.

Evaluated on the full 150-question FinanceBench benchmark, FinMA-RAG improves GPT-4o-judged accuracy from 19.3% to 44.0%, highlighting the importance of coupling ingestion, retrieval, and grounded reasoning rather than optimizing them in isolation.

Contributions. Our work makes three contributions: (1) We identify core failure modes in financial long-document QA, including structural fragmentation, terminology drift, and implicit numerical reasoning errors. (2) We propose FinMA-RAG, a unified architecture that integrates structure-aware ingestion, hybrid retrieval, and tool-augmented multi-agent reasoning. (3) We provide the first comprehensive end-to-end evaluation on FinanceBench demonstrating that retrieval-reasoning synergy yields more than a two-fold improvement over a vanilla RAG baseline.

2 Background and Problem

SEC 10-K and 10-Q filings present structural and reasoning challenges that go beyond those found in typical long-document QA tasks. These documents span hundreds of pages and interleave narrative analysis with dense financial statements, multi-

period comparisons, and irregular layouts. Answering questions over such filings typically requires locating dispersed evidence, extracting numeric values from tables, aligning fiscal periods, and performing arithmetic or comparative reasoning.

These requirements expose the limitations of conventional LLM and RAG pipelines. Single-step LLM generation frequently hallucinates unsupported numbers or confuses related accounting concepts, especially when numerical interpretation depends on subtle contextual cues. Naive retrieval pipelines suffer from table fragmentation caused by fixed-size chunking, dense retrievers retrieve semantically similar but fiscally incorrect passages, and sparse retrievers fail under issuer-specific terminology drift. Improvements in raw retrieval recall do not reliably translate into more accurate final answers when numerical grounding remains implicit.

Together, these challenges motivate a more structured approach to financial QA: one that preserves document topology during ingestion, integrates complementary retrieval signals to handle both semantic and lexical variability, and decomposes numerical reasoning into explicit, verifiable computation. FinMA-RAG operationalizes these principles through structure-preserving ingestion, hybrid multi-stage retrieval, and a multi-agent workflow with deterministic arithmetic.

3 Related Work

Financial Question Answering. Existing financial QA benchmarks, including FinQA (Chen et al., 2021), TAT-QA (Zhu et al., 2021), and ConvFinQA (Chen et al., 2022), focus on reasoning over short passages or pre-extracted tables. FinanceBench (Islam et al., 2023) extends this setting to full 10-K filings, where retrieval errors and numerical mistakes dominate overall failures. While prior systems emphasize numerical reasoning or table extraction, they commonly assume curated evidence. Our work instead targets the end-to-end challenges posed by full-length SEC filings, including scalable ingestion, robust retrieval under terminology drift, and verifiable multi-step arithmetic.

Document Ingestion and Long-Document Processing. Research on long-context modeling includes hierarchical attention (Beltagy et al., 2020; Zaheer et al., 2021), sparse attention (Tay et al., 2020), and structured PDF parsing (Rausch et al., 2021). These approaches improve text accessibility

but do not address the highly irregular, table-heavy layouts of SEC filings. Industrial pipelines often rely on standardized templates, whereas financial filings vary significantly across issuers and years. Our ingestion pipeline contributes a practical, reproducible method for preserving table structure while producing semantically coherent retrieval units at SEC scale.

Retrieval and Query Expansion. Dense retrieval (Karpukhin et al., 2020) and late-interaction models such as ColBERT (Santhanam et al., 2022) are widely used in open-domain QA. Hybrid sparse-dense retrieval, shown to improve robustness under terminology mismatch (Zhan et al., 2021; Formal et al., 2021a; Gao and Callan, 2021), is particularly relevant for financial filings, where issuer-specific vocabulary varies widely. Recent LLM-based query expansion methods (Nogueira and Lin, 2019; Wang et al., 2023) further mitigate lexical mismatch. Our retrieval module integrates these ideas to handle issuer terminology drift and to reliably surface fiscal-period-specific financial statements.

Reranking and Multi-Stage Retrieval. Multi-stage retrieval pipelines combining BM25, dense retrievers, and cross-encoder rerankers (Nogueira and Cho, 2019; Hofstätter et al., 2021; Luan et al., 2021) have become standard in open-domain QA. ColBERT v2 (Santhanam et al., 2022) provides efficient token-level late interaction for large candidate pools. We leverage late-interaction reranking to resolve fine-grained distinctions in financial evidence, particularly multi-year tables, where surface lexical similarity is high and dense retrieval alone is insufficient.

LLM Agents and Tool-Augmented Reasoning. Tool-augmented LLMs (Schick et al., 2023) and state-machine frameworks such as LangGraph enable structured, multi-step reasoning pipelines. However, their use in financial QA remains largely unexplored, despite known weaknesses of LLMs in quantitative reasoning (Lewkowycz et al., 2022a). Our work applies agent-based decomposition to explicitly isolate arithmetic into a deterministic calculator, ensuring numerical fidelity in financial settings.

Evaluation and Faithfulness. Beyond accuracy, recent metrics such as RAGAS (Es et al., 2025), TRUE (Honovich et al., 2022), and LLM-as-judge frameworks (Zheng et al., 2023) assess grounding

and faithfulness. FinanceBench further emphasizes numeric correctness. Our evaluation follows this trend by jointly measuring retrieval quality, grounding fidelity, and arithmetic accuracy.

4 Methods

FinMA-RAG is designed around three principles for reliable financial QA: preserving document structure during ingestion, maximizing recall and specificity through hybrid retrieval, and enforcing verifiable multi-step reasoning. The system consists of (i) a structure-aware ingestion pipeline, (ii) a hybrid sparse-dense retrieval module with late-interaction reranking, and (iii) a multi-agent workflow that performs explicit quantitative analysis.

4.1 Document Ingestion and Normalization

SEC filings exhibit substantial structural variability, including multi-row tables, cross-referenced statements, and issuer-specific formatting. To convert raw PDFs into retrieval-ready units, we use a layout-preserving extraction process that maintains line breaks, indentation patterns, and table boundaries. The extracted text is segmented using canonical SEC section headers (e.g., Item 7, Item 8), which provide consistent top-level structure across firms and years.

Within each section, narrative regions are split into semantically coherent spans while tabular blocks are kept intact by detecting alignment and repetition patterns. This strategy avoids fragmenting financial statements and preserves the local context needed for accurate numerical interpretation. Each chunk is embedded using the `intfloat/e5-large-v2` model—which showed the strongest domain performance in our retrieval experiments—and stored in a FAISS index along with metadata that supports efficient retrieval over multi-year corpora.

4.2 Hybrid Retrieval and Reranking

Financial questions require both broad coverage and precise selection of period-specific evidence. FinMA-RAG therefore adopts a hybrid retrieval strategy that integrates semantic and lexical signals. A Retriever Agent first expands each query into a set of structured sub-queries incorporating issuer identifiers, relevant SEC terminology, fiscal years, and domain-specific synonyms. These queries are issued simultaneously to FAISS for dense retrieval and to BM25 for sparse retrieval.

Dense retrieval captures paraphrased descriptions and contextual cues common in management discussions, while sparse retrieval ensures exact matching of line-item names and table headers. Retrieved candidates are merged and deduplicated to form a high-recall evidence pool.

To refine this pool, we apply ColBERT v2 as a late-interaction reranker. Its token-level matching is particularly effective for distinguishing fine-grained accounting expressions (e.g., operating income vs. operating expenses) and for resolving near-duplicate statements across fiscal years. The reranked passages constitute a compact, high-precision evidence set for downstream reasoning.

4.3 Multi-Agent Reasoning with Verifiable Arithmetic

Financial QA requires coordinated multi-step reasoning: identifying relevant disclosures, extracting numeric values, selecting fiscal periods, and performing deterministic arithmetic. FinMA-RAG enforces this workflow through four specialized agents operating within a LangGraph state machine:

- **Retriever Agent:** performs hybrid multi-query search and assembles reranked evidence.
- **Analyst Agent:** extracts quantities, interprets financial context, and determines whether computation is required.
- **Calculator Tool:** executes all arithmetic in a sandboxed, deterministic environment.
- **Synthesis Agent:** integrates retrieved evidence and computed values into a grounded final answer.

The Analyst Agent triggers tool calls whenever numerical reasoning is needed; arithmetic is never performed by the LLM. Returned values are passed back to the Analyst and Synthesis Agents, ensuring that each numerical statement in the final answer is directly linked to retrieved evidence and explicit computation. This decomposition mirrors professional financial workflows and significantly reduces reasoning drift.

4.4 Evaluation Protocol

We evaluate FinMA-RAG on the full FinanceBench benchmark using a GPT-4o judge focused on factual and numerical correctness. Re-

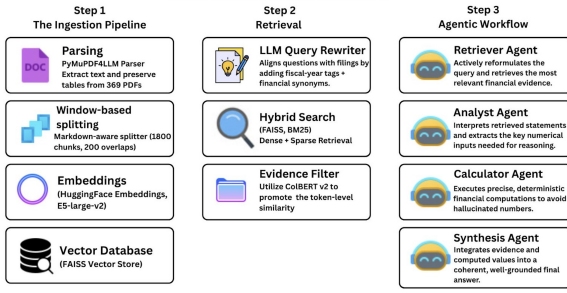


Figure 1: Overview of the FinMA-RAG architecture with structure-aware ingestion, hybrid retrieval, and tool-based multi-agent reasoning.

trieval performance is measured using recall, precision, and mean reciprocal rank, while grounding quality is assessed using RAGAS faithfulness, context precision, and answer correctness. For controlled ablations, we implement a vanilla RAG baseline with fixed-size chunking, dense-only retrieval without query expansion, and direct generation without reranking or tool usage, isolating the impact of each system component.

5 Experiments and Design Decisions

We adopt an iterative refinement approach, starting from a vanilla RAG baseline and incrementally incorporating retrieval and orchestration improvements based on error analysis of the FinanceBench dataset. All experiments are conducted on the full 150-question benchmark. For evaluation, we use GPT-4o as an automatic judge to assess factual correctness and numerical accuracy. Given the length and variability of valid answers, standard metrics such as Exact Match (EM), BLUE and ROUGE are insufficient. GPT-4o provides a scalable and semantically robust alternative, capable of evaluating meaning equivalence beyond surface form. We focus on evaluating retriever performance using Recall@5 and Mean Reciprocal Rank (MRR), and overall system performance using GPT-4o-judged accuracy. This evaluation setup enables rapid iteration and informed adjustment of experimental configurations.

5.1 Retrieval Experiments

High-quality retrieval is critical for the performance of retrieval-augmented generation systems, particularly in domains like finance where contextual relevance and precision are essential. To improve retrieval effectiveness, we conduct a series of targeted modifications: (1) evaluating multiple

open-source embedding models, (2) improving document ingestion, (3) addressing terminology drift via query expansion and hybrid retrieval, and (4) introducing a precision-oriented reranking stage. Each modification is guided by concrete failure cases observed in the baseline system.

5.1.1 Impact of Embedding Model on Retrieval Quality

To assess the contribution of the embedding model to retrieval performance, we evaluated three state-of-the-art dense encoders, BAAI/BGE-BASE-EN-V1.5 (768-dimensional), BAAI/BGE-LARGE-EN-V1.5 (1024-dimensional), and INTFLOAT/E5-LARGE-V2 (1024-dimensional), on the FinanceBench retrieval task. The BGE models share the same architecture but differ in size and output dimensionality, allowing us to isolate the effect of representation capacity. In contrast, E5-large-v2 is instruction-tuned specifically for retrieval and produces high-dimensional embeddings optimized for semantic relevance. Each model was applied to an identical corpus of markdown-preserved document chunks with a chunk size of approximately 1800 tokens and 200-token overlap, and retrieval was performed using FAISS without any reranking. We report Recall@5 and Mean Reciprocal Rank (MRR) as primary metrics.

As shown in Figure 2, retrieval performance improves consistently with model capacity: Recall@5 increases from 0.53 $\rightarrow$ 0.59 $\rightarrow$ 0.76, and MRR increases from 0.31 $\rightarrow$ 0.35 $\rightarrow$ 0.54 across the three encoders. These gains suggest that larger embedding models are better able to encode fine-grained financial semantics, particularly distinctions tied to fiscal years, accounting structures, and line-item details. Based on this evaluation, we adopt INTFLOAT/E5-LARGE-V2 for all subsequent experiments, as it provides the strongest semantic representations and establishes a more reliable retrieval foundation for our hybrid system.

5.1.2 Improving Document Ingestion with Structure-Preserving Extraction

Our initial error analysis revealed that many retrieval failures were rooted in structural artifacts introduced during PDF ingestion. Standard PyMuPDF extraction often fragmented balance sheets, split multi-row tables, and produced chunks with misaligned numerical columns. These issues disrupted the semantic and structural integrity of

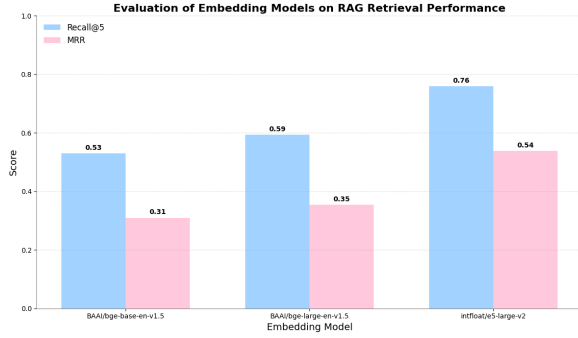


Figure 2: Evaluation of Embedding Models on RAG Retrieval Performance.

financial statements, resulting in incomplete or misleading retrieval candidates.

To mitigate this, we adopted PyMuPDF4LLM, an enhanced extraction framework that preserves layout and table structure by converting filings into markdown with stable row boundaries and hierarchical formatting. This structure-preserving approach substantially improved retrieval coherence: financial tables were retained as unified retrieval units, narrative text was cleanly separated, and statements no longer spanned multiple disjointed chunks.

The improved ingestion pipeline enabled the retriever to access more complete and contextually grounded evidence, significantly strengthening the reliability of downstream retrieval and reasoning components.

5.1.3 Enhancing Recall with Query Expansion and Hybrid Sparse–Dense Retrieval

Despite improvements in chunk quality, dense retrieval alone often failed to surface the correct evidence due to issuer-specific terminology drift. For example, FinanceBench queries frequently reference “revenue,” whereas filings may use “net sales,” “operating revenues,” or domain-specific equivalents. Dense embeddings capture semantic similarity but struggle with these lexical variations. To mitigate the mismatch, we introduced LLM-driven query expansion in which GPT-4o rewrote user queries into multiple SEC-style paragraphs while enforcing fiscal-year constraints. We further adopted a hybrid retrieval system that merged FAISS dense search with BM25 lexical retrieval. Prior research demonstrates that hybrid sparse–dense retrieval is particularly effective in domains with high terminological variability (Formal et al., 2021b; Ma et al., 2024). Empirically, this approach substantially improved recall: correct-

year financial statements surfaced more frequently, and the retrieval of irrelevant narrative sections decreased.

5.1.4 Increasing Precision with ColBERT v2 Reranking

While hybrid retrieval substantially improved the likelihood that relevant financial statements appeared within the candidate set, the raw top-k results often ranked semantically related but ultimately irrelevant segments, such as Management Discussion and Analysis (MD&A) narratives or fiscal data from incorrect reporting periods—above the gold evidence. To address this, we applied ColBERT v2 (Santhanam et al., 2022) as a late-interaction reranker over the merged FAISS and BM25 outputs. By computing fine-grained token-level similarity scores, ColBERT effectively distinguishes near-duplicate passages and reliably surfaces table rows that precisely match the user query (Khattab and Zaharia, 2020; Santhanam et al., 2022).

Notably, retrieval metrics such as Recall@5 and MRR are reported before reranking; ColBERT does not influence recall, but improves candidate ordering. This reordering proved especially impactful for multi-year comparison questions, where correctly identifying the fiscal period is critical. By consistently promoting gold evidence to higher ranks, ColBERT improved the precision of retrieved contexts, enabling more accurate numerical extraction and contributing significantly to the overall gains in FinanceBench accuracy.

5.2 Multi-Agent Orchestration Experiments

Despite improved retrieval quality, the vanilla RAG system continued to produce incorrect answers, primarily due to arithmetic hallucinations and unreliable numerical reasoning. Qualitative analysis revealed common failure modes, including incorrect percentage calculations, erroneous year-over-year comparisons, and digit hallucinations when interpreting tables, patterns consistent with prior work highlighting the limitations of large language models (LLMs) on quantitative tasks (Lewkowycz et al., 2022b). To address these issues, we introduced a deterministic calculator module to handle arithmetic operations. The LLM was restricted to extracting operands from the retrieved context and delegating computation to the external calculator. This separation of concerns reliably eliminated arithmetic errors and produced consistent, verifi-

able results.

We then integrated retrieval, reasoning, computation, and synthesis into a multi-agent workflow implemented with LangGraph. The retriever agent managed query expansion and multi-stage retrieval; the analyst agent identified relevant numerical values; the calculator executed all arithmetic; and the synthesis agent generated the final grounded answer. This modular orchestration reduced cognitive load on the LLM, prevented chain-of-thought drift, and ensured alignment between evidence and computation.

6 Results and Analysis

With the full multi-agent pipeline, the system achieves an end-to-end accuracy of 44%, highlighting both progress and the remaining challenges in financial question answering. Several factors contribute to this performance ceiling. A primary bottleneck lies in retrieval quality: gains in recall, precision, and MRR are necessary to ensure that relevant evidence is consistently surfaced for downstream reasoning. Even small retrieval errors can have disproportionate effects in financial QA, where missing a single line item or fiscal reference may invalidate the entire answer. Second, although the multi-agent architecture introduces modularity and task decomposition, and includes mechanisms to reduce error propagation, it remains susceptible to cascading failures in practice. Errors in early components, particularly retrieval or entity extraction, can still propagate downstream, as later agents often operate under the assumption that upstream outputs are correct. This silent propagation of flawed inputs reduces overall system robustness, especially in tasks requiring tightly coupled reasoning over numerical and temporal evidence. Third, LLMs continue to exhibit limitations in numerical and symbolic reasoning. Despite having access to correct evidence, models may still produce outputs that are fluent but mathematically or logically incorrect, highlighting the need for more grounded and verifiable reasoning mechanisms in high-stakes domains like finance.

Our results also highlight key insights into how retrieval quality and structured multi-agent reasoning jointly affect performance in financial question answering. First, the choice of embedding model significantly influences retrieval effectiveness. We observe a steady improvement in Recall@5, from 0.53 with BGE-base to 0.59 with BGE-large and

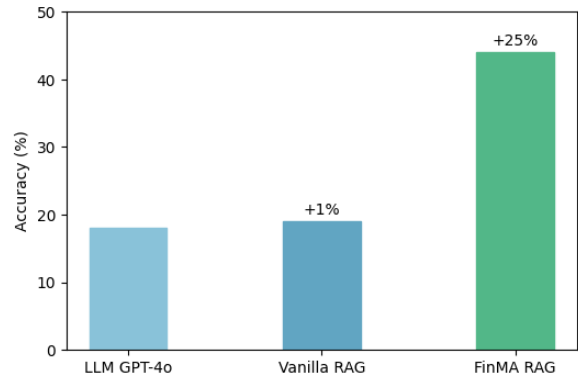


Figure 3: Evaluation of Embedding Models on RAG Retrieval Performance.

0.76 with E5-large-v2, highlighting the importance of embedding model choice in retrieval performance. While the absolute gains are moderate, they meaningfully increase the likelihood that the correct evidence is retrieved, laying a stronger foundation for downstream reasoning. Although we experimented with various chunking strategies, including adjusting chunk size and overlap, as well as using a dynamic chunker conditioned on document context, the improvements were marginal compared to those achieved by stronger embedding models. These findings suggest that embedding quality plays a more critical role than segmentation strategy in retrieval-centric financial QA, where semantic precision and factual grounding are essential. This indicates that domain-specific embedding models may offer substantial advantages over general-purpose embeddings in specialized domains (Tang and Yang, 2025).

Second, document parsing quality plays a similarly critical role. Naive PDF extraction methods often fail to preserve the structural integrity of financial tables, resulting in missing numerical values and fragmented evidence. In contrast, layout-aware parsing with PyMuPDF4LLM restores Markdown structure and recovers tabular content, improving both evidence completeness and the reliability of downstream reasoning.

Third, hybrid retrieval combined with ColBERT v2 reranking enhances retrieval precision not by significantly altering recall, but by improving the ordering of candidate passages. Prioritizing fact-dense and contextually relevant chunks at the top of the ranked list substantially improves the LLM’s ability to ground its responses, even when the underlying candidate set remains unchanged.

Fourth, the multi-agent architecture mitigates

several common LLM failure modes. The Analyst Agent ensures structured extraction of line items and fiscal periods; the Calculator Tool performs explicit arithmetic to eliminate numeric hallucinations; and the Synthesis Agent integrates retrieved evidence while verifying consistency across intermediate steps. This modular decomposition reduces the cognitive burden on any single agent and increases robustness on complex, multi-step financial queries.

Finally, LangGraph’s multi-step control flow enables iterative verification and recomputation, supporting a level of reasoning consistency that is rarely attainable with single-pass LLMs. This iterative mechanism allows the system to refine intermediate outputs, correct earlier extraction errors, and enforce alignment between retrieved evidence and final computations.

A central finding of this work is that even modest improvements in retrieval recall can lead to substantial gains in end-to-end accuracy when combined with structured reasoning and tool-based augmentation. In the context of financial question answering, performance depends not only on retrieving relevant information, but on the system’s ability to interpret and apply that information in a controlled and verifiable manner.

7 Limitations

Content-Aware Chunking. Financial filings exhibit heterogeneous structural formats, including tables, footnotes, and mixed narrative-tabular sections, which often misalign with fixed-size chunking schemes. Naive segmentation strategies can fragment table rows, separate headers from their corresponding values, or disrupt semantically coherent blocks, ultimately degrading the quality and completeness of retrieved evidence. Designing content-aware chunking methods that preserve structural and semantic integrity remains an open challenge. While our approach employed fixed-length chunks with overlapping windows, future work may benefit from adaptive segmentation strategies that align more closely with document layout and discourse boundaries.

Hybrid Retrieval Integration. While sparse (BM25) and dense (FAISS) retrievers offer complementary strengths, combining their outputs in a robust and query-sensitive manner remains challenging. Improper weighting can lead to redundancy, elevate semantically similar but irrelevant passages,

or suppress critical evidence. This issue is further compounded in financial QA, where query formulations vary widely in lexical specificity and semantic scope. Developing principled fusion methods, potentially adaptive or query-aware, remains an open direction for realizing the full potential of hybrid retrieval in domain-specific settings.

Single Embedding Store. The current architecture relies on a single unified embedding index, which may limit retrieval diversity and reduce coverage across heterogeneous document structures. Financial filings often contain semantically distinct content types, such as narratives, tables, and footnotes, that may benefit from separate representation spaces. Incorporating multiple specialized embedding stores tailored to structural categories could expand the effective search space, improve recall, and enhance robustness on complex, multi-format financial queries.

8 Future Directions

Format-Aware and Adaptive Chunking. Overcoming the limitations of fixed-size segmentation will require chunking strategies that explicitly account for document structure. Promising directions include layout-aware chunkers, table-specific parsers, and multimodal models that incorporate textual, structural, and visual cues. Learning-based or adaptive approaches that dynamically adjust chunk boundaries based on content type may further improve semantic coherence, particularly in table-centric financial documents.

Reducing Embedding Redundancy and Noise. To address the similarity saturation observed in dense embedding spaces, future work may explore deduplication pipelines, hierarchical clustering of embedding neighborhoods, or density-aware indexing techniques. Additionally, methods such as semantic hashing, embedding compression, and hybrid coarse-to-fine retrieval may improve embedding discriminability while maintaining high recall, ultimately enhancing retrieval precision in redundant documents.

More Principled Hybrid Retrieval. Integrating sparse and dense retrieval in a robust and query-sensitive manner remains a core challenge in retrieval-augmented systems. Promising directions include learning-to-rank fusion models, query-adaptive weighting strategies, and retrieval policies optimized through reinforcement learning.

Developing more dynamic and interpretable fusion mechanisms that effectively combine lexical and semantic signals may improve evidence selection across diverse query types, particularly in domains with high linguistic and structural variability.

Multiple Specialized Embedding Stores. Extending beyond a single embedding store presents a promising direction for improving retrieval coverage in structurally diverse documents. Future systems may benefit from constructing separate embedding indexes for distinct content types, such as narratives, tables, footnotes, or computed financial indicators, and dynamically selecting or aggregating results based on query semantics. This specialization can be further enhanced by incorporating domain-specific embedding models tuned for financial text, which may improve representation quality for content with highly specialized vocabulary and structure. Together, these advances could increase retrieval precision and robustness in domain-specific QA settings.

Strengthening Numerical and Symbolic Reasoning. The persistent difficulty of LLMs with arithmetic and symbolic manipulation underscores the need for more robust numerical grounding. Future architectures may incorporate external solvers, differentiable calculators, or program-induction components to support explicit symbolic reasoning rather than relying on latent approximations. Another promising direction is the development of joint training or co-execution frameworks that more tightly couple retrieved evidence with computational procedures, ensuring that numerical operations remain faithful to source data and are verifiable.

9 Conclusion

We introduced FinMA-RAG, a domain-adaptive RAG framework designed to address the structural irregularities, lexical variation, and numerical complexity that make financial QA over full SEC filings uniquely challenging. By combining structure-preserving ingestion, hybrid sparse-dense retrieval with late-interaction reranking, and a multi-agent workflow with explicit tool-based arithmetic, FinMA-RAG substantially improves answer accuracy on the full FinanceBench benchmark. Controlled ablations confirm that each component contributes distinct gains in evidence fidelity and numerical correctness, underscoring the importance

of system-level design rather than single-module optimization.

Beyond empirical improvements, our results reveal broader lessons for financial QA. Retrieval quality depends critically on structural normalization and terminology alignment, not just embedding strength. Late-interaction reranking improves passage specificity even when recall is unchanged, and reliable numerical reasoning requires explicit decomposition into verifiable operations. These patterns highlight that long-context financial QA hinges on tightly coupling ingestion, retrieval, and reasoning in a coordinated pipeline.

Despite more than doubling performance over a vanilla RAG baseline, significant challenges remain. Dense retrieval continues to suffer from similarity saturation, domain-agnostic embeddings struggle with fine-grained financial semantics, and hybrid retrieval remains sensitive to query formulation and calibrating sparse-dense contributions. Moreover, tool-augmented reasoning currently handles only limited compositional operations. As a result, system performance remains capped by structural noise in SEC filings, ambiguity in financial terminology, and retrieval saturation in dense embedding spaces, all of which limit reasoning fidelity even under multi-agent orchestration.

Future progress will likely require adaptive chunking, multi-store or table-aware indexing, and richer symbolic interfaces capable of supporting more complex financial analyses. More broadly, our findings emphasize that high-stakes domains such as finance demand retrieval and reasoning architectures that are not only accurate, but also interpretable and auditable.

References

- Iz Beltagy, Matthew E. Peters, and Arman Cohan. 2020. [Longformer: The long-document transformer](#). *Preprint*, arXiv:2004.05150.
- Zhiyu Chen, Wenhui Chen, Charese Smiley, Sameena Shah, Iana Borova, Dylan Langdon, Reema Moussa, Matt Beane, Ting-Hao Huang, Bryan Routledge, and William Yang Wang. 2021. [FinQA: A dataset of numerical reasoning over financial data](#). In *Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing*, pages 3697–3711, Online and Punta Cana, Dominican Republic. Association for Computational Linguistics.
- Zhiyu Chen, Shiyang Li, Charese Smiley, Zhiqiang Ma, Sameena Shah, and William Yang Wang. 2022. [Confinqa: Exploring the chain of numerical reasoning in](#)

- conversational finance question answering. *Preprint*, arXiv:2210.03849.
- Shahul Es, Jithin James, Luis Espinosa-Anke, and Steven Schockaert. 2025. [Ragas: Automated evaluation of retrieval augmented generation](#). *Preprint*, arXiv:2309.15217.
- Mathieu Formal, Carlos Lassance, Benjamin Piwowarski, and Stéphane Clinchant. 2021a. Splade v2: Sparse lexical and expansion model for information retrieval. In *NeurIPS Workshop on Information Retrieval*.
- Tristan Formal, Carlos Lassance, Benjamin Piwowarski, and Stéphane Clinchant. 2021b. A comparative study on dense and sparse representations for passage retrieval. In *Findings of the Association for Computational Linguistics: EMNLP 2021*, pages 2709–2720.
- Luyu Gao and Jamie Callan. 2021. A document expansion model for passage retrieval. In *ICTIR*.
- Sebastian Hofstätter, Sheng-Chieh Lin, Jheng-Hong Yang, Jimmy Lin, and Allan Hanbury. 2021. [Efficiently teaching an effective dense retriever with balanced topic aware sampling](#). *Preprint*, arXiv:2104.06967.
- Or Honovich, Roei Aharoni, Jonathan Herzig, Hagai Taitelbaum, Doron Kukliansky, Vered Cohen, Thomas Scialom, Idan Szpektor, Avinatan Hassidim, and Yossi Matias. 2022. [True: Re-evaluating factual consistency evaluation](#). *Preprint*, arXiv:2204.04991.
- Pranab Islam, Anand Kannappan, Douwe Kiela, Rebecca Qian, Nino Scherrer, and Bertie Vidgen. 2023. [Financebench: A new benchmark for financial question answering](#). *arXiv preprint arXiv:2311.11944*.
- Vladimir Karpukhin, Barlas Oguz, Sewon Min, Patrick Lewis, Ledell Wu, Sergey Edunov, Danqi Chen, and Wen-tau Yih. 2020. [Dense passage retrieval for open-domain question answering](#). In *EMNLP*.
- Omar Khattab and Matei Zaharia. 2020. [ColBERT: Efficient and effective passage search via contextualized late interaction over BERT](#). In *SIGIR*.
- Aitor Lewkowycz, Anders Andreassen, David Dohan, Ethan Dyer, Henryk Michalewski, Vinay Ramasesh, Ambrose Slone, Cem Anil, Imanol Schlag, Theo Gutman-Solo, Yuhuai Wu, Behnam Neyshabur, Guy Gur-Ari, and Vedant Misra. 2022a. [Solving quantitative reasoning problems with language models](#). *Preprint*, arXiv:2206.14858.
- Aitor Lewkowycz, David Dohan, Oscar Sainz, Nora Schucher, Anna Mishchenko, Jacob Austin, Olivier Bousquet, Quoc V. Le, and Slav Petrov. 2022b. Solving quantitative reasoning problems with language models. In *Advances in Neural Information Processing Systems*.
- Yi Luan, Jacob Eisenstein, Kristina Toutanova, and Michael Collins. 2021. [Sparse, dense, and attentional representations for text retrieval](#). *Preprint*, arXiv:2005.00181.
- Xiaojun Ma, Yixin Liu, and Wei Zhao. 2024. Query expansion with large language models for domain-specific retrieval. *arXiv preprint arXiv:2403.01700*.
- Rodrigo Nogueira and Kyunghyun Cho. 2019. [Passage re-ranking with BERT](#). *arXiv preprint arXiv:1901.04085*.
- Rodrigo Nogueira and Jimmy Lin. 2019. From doc2query to DocTTTTTQuery. In *EMNLP Workshop on Neural Information Retrieval*.
- Johannes Rausch, Octavio Martinez, Fabian Bissig, Ce Zhang, and Stefan Feuerriegel. 2021. [Docparser: Hierarchical structure parsing of document renderings](#). *Preprint*, arXiv:1911.01702.
- Keshav Santhanam, Daniel Campos, Ronak Pradeep, Xiaohui Ma, Rodrigo Nogueira, and Jimmy Lin. 2022. Colbertv2: Effective and efficient retrieval via lightweight late interaction. *arXiv preprint arXiv:2112.01488*.
- Timo Schick, Jane Dwivedi-Yu, Roberto Dessì, Roberta Raileanu, Maria Lomeli, Luke Zettlemoyer, Nicola Cancedda, and Thomas Scialom. 2023. [Toolformer: Language models can teach themselves to use tools](#). *Preprint*, arXiv:2302.04761.
- Yixuan Tang and Yi Yang. 2025. [Finmtb: Finance massive text embedding benchmark](#). In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pages 3620–3638.
- Yi Tay, Dara Bahri, Liu Yang, Donald Metzler, and Da-Cheng Juan. 2020. [Sparse sinkhorn attention](#). *Preprint*, arXiv:2002.11296.
- Xinyu Wang, Changlong Yu, Xiaohui Liu, Yinqing Chen, Jiakai Zhang, Zihan Liu, Weizhe Wen, Yuheng Yang, Da Yin, and Ming Tan. 2023. Query expansion by large language models. *arXiv preprint arXiv:2305.14750*.
- Manzil Zaheer, Guru Guruganesh, Avinava Dubey, Joshua Ainslie, Chris Alberti, Santiago Ontanon, Philip Pham, Anirudh Ravula, Qifan Wang, Li Yang, and Amr Ahmed. 2021. [Big bird: Transformers for longer sequences](#). *Preprint*, arXiv:2007.14062.
- Zhuosheng Zhan, Puxuan Mao, Weize Yao, Zihao Zheng, Shaohan Huang, Fei Huang, Enhong Chen, and Shikun Zhang. 2021. Coil: Revisit exact lexical match in information retrieval with contextualized inverted list. In *ACL*.
- Lianmin Zheng, Wei-Lin Chiang, Ying Sheng, Siyuan Zhuang, Zhanghao Wu, Yonghao Zhuang, Zi Lin, Zhuohan Li, Dacheng Li, Eric P. Xing, Hao Zhang, Joseph E. Gonzalez, and Ion Stoica. 2023. [Judging llm-as-a-judge with mt-bench and chatbot arena](#). *Preprint*, arXiv:2306.05685.

Fengbin Zhu, Wenqiang Lei, Youcheng Huang, Chao Wang, Shuo Zhang, Jiancheng Lv, Fuli Feng, and Tat-Seng Chua. 2021. [Tat-qa: A question answering benchmark on a hybrid of tabular and textual content in finance](#). *Preprint*, arXiv:2105.07624.